Appendix A: Summary of Experimental Conditions

Table 1 describes the 53 experimental conditions included in the model comparison, including the number of trials, the number of words presented per trial (Words/Trial), the number of referents presented per trial (Objects/Trial), the number of to-be-learned word-referent pairs (Items), people's overall mean accuracy ($p(o|w)$ across all intended $w-o$ mappings) in each condition (Accuracy), the standard deviation of performance (SD), and the number of participants per condition (N).

Appendix B: Clustering experiments and models by misfit

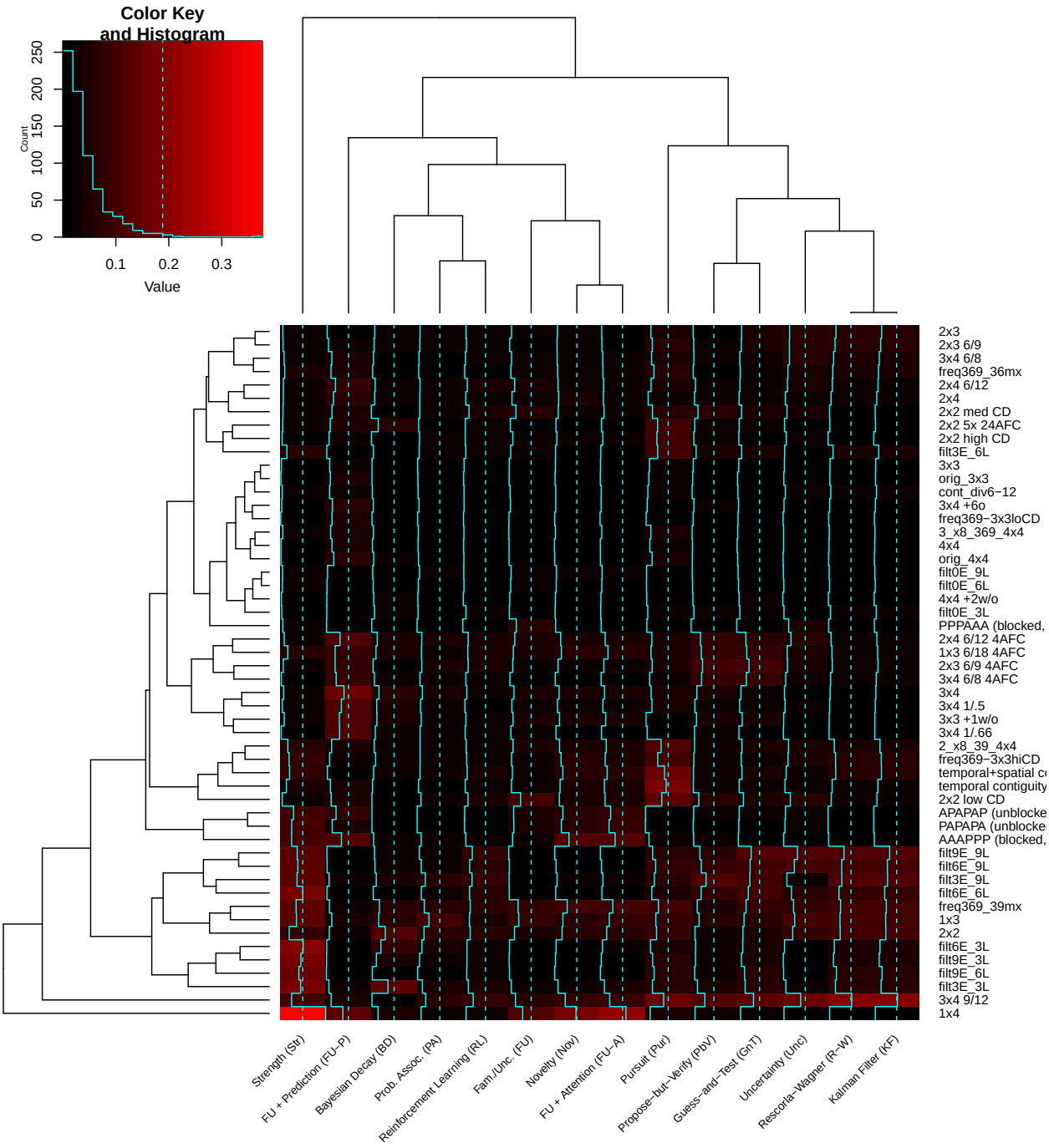
The heatmap below shows cross-validated model fit (SSE; sum of squared error) for each experimental condition, across all 14 fitted models and the 52 conditions with cross-validated fits. Note that each model has difficulty fitting at least one or more experimental conditions–and these difficult conditions vary somewhat by model. For example, the strength-biased model has particular difficulty with the `filt` conditions[1] (Kachergis et al., 2012), which present a group of word-referent pairs early in training which then systematically co-occur with particular novel late-stage word-referent pairs, testing how strictly learners will maintain a mutual exclusivity (ME) constraint. The Propose-but-Verify model also shows greater misfit in most of these conditions (except for the `filtXE_3L` conditions, which have only 3 repetitions of the late-stage pairings, and thus do not overwhelm learners' ME bias).

On the other hand, many experimental conditions are nearly equally well fit by all models, especially those that have a fixed number of repetitions per word-referent pair (e.g., 3x3 and 4x4, although 2x2 presents difficulties for some models).

[1] Except for the `filtOE_` conditions, which consist only of the late-stage pairs, with no early stage.

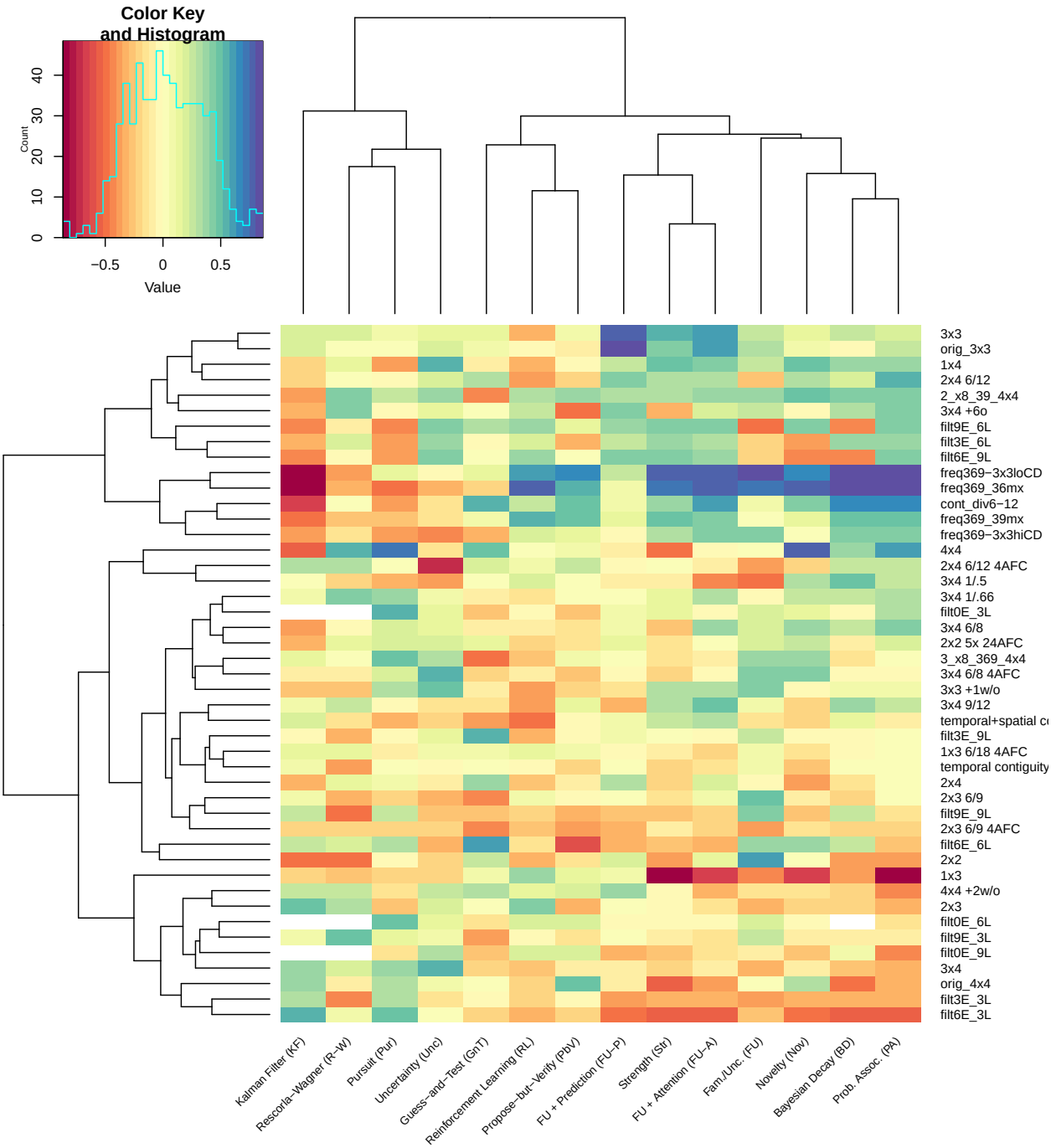
Condition Name	Trials	Words/Trial	Objects/Trial	Items	Accuracy	SD	N
3x4	36	3	4	18	0.19	0.12	25.00
3x4 1/.5	36	3	4	18	0.22	0.11	25.00
3x4 1/.66	36	3	4	18	0.21	0.09	25.00
3x4 +6o	36	3	4	18	0.27	0.12	20.00
2x4	54	2	4	18	0.30	0.15	33.00
3x3 +1w/o	54	3	3	18	0.17	0.08	39.00
4x4 +2w/o	54	4	4	18	0.10	0.05	39.00
1x3 6/18 4AFC	108	1	3	18	0.67	0.10	43.00
2x3 6/9 4AFC	54	2	3	18	0.69	0.11	38.00
2x4 6/12 4AFC	54	2	4	18	0.62	0.12	31.00
3x4 6/8 4AFC	36	3	4	18	0.69	0.09	36.00
1x3	108	1	3	18	0.74	0.11	23.00
2x3	54	2	3	18	0.58	0.07	23.00
2x4 6/12	54	2	4	18	0.33	0.18	14.00
3x4 6/8	36	3	4	18	0.42	0.14	13.00
2x3 6/9	54	2	3	18	0.55	0.11	32.00
3x4 9/12	54	3	4	18	0.69	0.08	33.00
1x4	108	1	4	18	0.19	0.10	40.00
1x3	108	1	3	18	0.39	0.12	40.00
4x4	27	4	4	18	0.31	0.07	77.00
3x3	36	3	3	18	0.43	0.08	36.00
2x2	54	2	2	18	0.79	0.11	19.00
2x2 5x 24AFC	60	2	2	24	0.51	0.12	46.00
filt0E_3L	18	2	2	12	0.38	0.09	31.00
filt3E_3L	27	2	2	12	0.72	0.12	30.00
filt6E_3L	36	2	2	12	0.71	0.09	30.00
filt9E_3L	45	2	2	12	0.70	0.08	31.00
filt0E_6L	36	2	2	12	0.47	0.11	27.00
filt3E_6L	45	2	2	12	0.67	0.11	27.00
filt6E_6L	54	2	2	12	0.79	0.09	27.00
filt9E_6L	63	2	2	12	0.75	0.09	27.00
filt0E_9L	54	2	2	12	0.54	0.09	31.00
filt3E_9L	63	2	2	12	0.83	0.08	31.00
filt6E_9L	72	2	2	12	0.82	0.09	31.00
filt9E_9L	81	2	2	12	0.86	0.06	31.00
2_x8_39_4x4	27	4	4	18	0.41	0.16	30.00
3_x8_369_4x4	27	4	4	18	0.33	0.06	74.00
freq369-3x3loCD	36	3	3	18	0.33	0.08	102.00
freq369-3x3hiCD	36	3	3	18	0.56	0.12	26.00
freq369_36mx	36	3	3	18	0.45	0.16	62.00
freq369_39mx	36	3	3	18	0.62	0.14	66.00
orig_4x4	27	4	4	18	0.27	0.07	88.00
orig_3x3	36	3	3	18	0.43	0.08	104.00
cont_div6-12	36	3	3	18	0.43	0.11	40.00
temporal+spatial contiguity	27	4	4	18	0.49	0.08	30.00
temporal contiguity only	27	4	4	18	0.40	0.10	32.00
2x2 low CD	16	2	2	8	0.34	0.00	84.00
2x2 med CD	16	2	2	8	0.39	0.00	84.00
2x2 high CD	16	2	2	8	0.48	0.00	84.00
AAAPPP (blocked, A-first)	48	1	4	8	0.18	0.00	64.00
APAPAP (unblocked, A-first)	48	1	4	8	0.23	0.00	64.00
PAPAPA (unblocked, P-first)	48	1	4	8	0.23	0.00	64.00
PPPAAA (blocked, P-first)	48	1	4	8	0.32	0.00	64.00

Table 1: Summary of modeled datasets.



Correlation of model vs. human item-level performance per condition

Each cell displays the correlation coefficient of model vs. human item-level performance in a given experimental condition. 7 conditions (the Koehne et al. 2013 and Suanda et al. 2014 conditions, which only have condition-level human accuracy) are excluded here, since a correlation is undefined without item-level variance in the human data; they are included in the SSE-based heatmap above and in Table 2 below.



Which experimental designs are easy or hard, overall?

The heatmaps above group misfit by model; here we instead average over models, to ask which experimental designs are uniformly easy or hard to predict for *any* model – a useful signal for choosing which conditions actually discriminate between theories, versus which conditions every model gets right (or wrong) regardless of mechanism.

Table 2 shows the five conditions with the lowest mean SSE across all 14 models (easiest, fit well by essentially every model) and the five with the highest (hardest, fit poorly by essentially every model). The `SD SSE` column shows how much models disagree on a condition: a low SD alongside a high mean SSE means every

model struggles with that design about equally – a design that is hard for principled reasons rather than one that merely discriminates between theories.

Model disagreement: a discriminability index independent of human data

Mean SSE (above) indexes how *hard* a condition is, but hardness and discriminative *value* are different things: a condition every model gets equally wrong is hard but uninformative for choosing between theories, while a condition that splits the models – whoever turns out to be right – is exactly the kind of experiment worth running again with more power, or in a new population. We index this directly, independent of human data, as the standard deviation of each model's predicted item-level accuracy (`ModelPerf`, at each model's group-fit parameters) across models, averaged over the items in a condition. Because several of our associative models are one-parameter-removed cousins of each other (FU/FUs/FU-A/FU-P/Str/Unc/Nov all share the same core mechanism), their mutual spread can pad this index without reflecting genuine disagreement between *theories*. We therefore report it two ways: across all 16 fitted models, and across 10 representatives, one per distinct mechanism (Fam./Unc. (FU), Prob. Assoc. (PA), Bayesian Decay (BD), Rescorla-Wagner (R-W), Propose-but-Verify (PbV), Guess-and-Test (GnT), Pursuit (Pur), Reinforcement Learning (RL), Kalman Filter (KF), Baseline) – the two versions correlate at 0.86, but reorder some conditions meaningfully (see below).

Table 3 shows the most- and least-discriminative conditions by this measure. The `filt9E_*`/`filt6E_*` conditions – which present a group of pairs early, then systematically pair a subset of the late-stage objects with the early-stage words to test how strictly a mutual-exclusivity bias is maintained – are the most model-discriminating conditions in the whole set, on both versions of the index; the temporal- and spatial-contiguity conditions also rise near the top once the FU family's internal spread is removed, indicating a genuine cross-theory split there rather than one associative model disagreeing with its own close cousins. At the other end, `filt0E_6L`/`filt0E_9L` (the late-stage-only `filt` conditions, with no early pairs to violate) and several of the fixed-repetition `3x4`/`4x4` conditions are answered almost identically by every model regardless of mechanism – useful as sanity checks, but uninformative for adjudicating between theories. This index correlates with the SSE-based "hardest" ranking above at only $r \approx 0.51$, confirming that "hard for everyone" and "discriminates between theories" are related but distinct properties of a design.

Each Model's Best- and Worst-Fitting Experiment

Table 4 summarizes for each model, what experimental condition is best fit by that model (in terms of correlation, `best` and `best_cond`), and what condition has the worst fit for that model (`worst` and `worst_cond`), again excluding the 7 conditions with no item-level human variance.

Across models, freq369-3x3loCD is most often the best-fit condition (best for 4 of 14 models), and 1x3 is most often the worst-fit condition (worst for 4 models).

Best- and Worst-Fitting Models per Experiment

Table 5 takes the complementary view: for each experimental condition, which model fits it best and which fits it worst (by item-level correlation), and then, aggregating over conditions, how many conditions each model "wins" (is the single best-fitting model for) and "loses" (is the single worst-fitting model for). A model that wins many, varied conditions is a safe default; a model that wins a few conditions no other model wins is doing something distinctive worth understanding.

Rank	Condition	Mean r	Mean SSE	SD SSE
Easiest	4x4 +2w/o	0.025	0.010	0.006
Easiest	freq369-3x3loCD	0.373	0.010	0.013
Easiest	orig_3x3	0.216	0.013	0.010
Easiest	4x4	0.177	0.013	0.014
Easiest	filt0E_6L	0.079	0.014	0.004
Hardest	3x4 9/12	0.053	0.106	0.055
Hardest	1x4	0.174	0.093	0.106
Hardest	freq369_39mx	0.152	0.085	0.028
Hardest	1x3	-0.290	0.082	0.023
Hardest	filt3E_9L	0.024	0.070	0.044

Table 2: The five easiest and five hardest experimental conditions, by mean SSE across all fitted models.

Rank	Condition	SD (all models)	SD (mechanism reps)
Most discriminative	filt9E_9L	0.171	0.163
Most discriminative	filt6E_9L	0.161	0.152
Most discriminative	filt9E_6L	0.161	0.150
Most discriminative	1x3 6/18 4AFC	0.154	0.161
Most discriminative	filt9E_3L	0.153	0.141
Most discriminative	filt6E_6L	0.149	0.136
Least discriminative	cont_div6-12	0.071	0.056
Least discriminative	4x4 +2w/o	0.065	0.066
Least discriminative	filt0E_6L	0.052	0.049
Least discriminative	filt0E_9L	0.040	0.037

Table 3: Most- and least-discriminative experimental conditions: standard deviation of model-predicted item accuracy across models, at each model's group-fit parameters. 'all models' is 16 models (the 15-model roster plus baseline); 'mechanism reps' is one representative per distinct theoretical mechanism.

Model	best	worst	best_cond	worst_cond
Bayesian Decay (BD)	0.87	-0.53	freq369-3x3loCD	filt6E_3L
Prob. Assoc. (PA)	0.85	-0.84	freq369_36mx	1x3
FU + Prediction (FU-P)	0.81	-0.47	orig_3x3	filt6E_3L
Fam./Unc. (FU)	0.81	-0.50	freq369-3x3loCD	filt9E_6L
FU + Attention (FU-A)	0.81	-0.64	freq369_36mx	1x3
Novelty (Nov)	0.78	-0.68	4x4	1x3
Reinforcement Learning (RL)	0.78	-0.46	freq369_36mx	temporal+spatial contiguity
Strength (Str)	0.76	-0.83	freq369-3x3loCD	1x3
Pursuit (Pur)	0.74	-0.48	4x4	freq369_36mx
Propose-but-Verify (PbV)	0.65	-0.58	freq369-3x3loCD	filt6E_6L
Guess-and-Test (GnT)	0.61	-0.48	filt6E_6L	3_x8_369_4x4
Uncertainty (Unc)	0.57	-0.69	3x4	2x4 6/12 4AFC
Kalman Filter (KF)	0.53	-0.86	filt6E_3L	freq369-3x3loCD
Rescorla-Wagner (R-W)	0.52	-0.48	4x4	filt9E_9L

Table 4: Each model's best- and worst-fitting experiment.

Model	best_fits	worst_fits
Uncertainty (Unc)	6	2
FU + Prediction (FU-P)	5	0
Prob. Assoc. (PA)	5	3
Fam./Unc. (FU)	4	4
Kalman Filter (KF)	4	8
Rescorla-Wagner (R-W)	4	4
Guess-and-Test (GnT)	3	7
Novelty (Nov)	3	3
Pursuit (Pur)	3	1
Bayesian Decay (BD)	2	0
FU + Attention (FU-A)	2	2
Reinforcement Learning (RL)	2	6
Propose-but-Verify (PbV)	1	4
Strength (Str)	1	1

Table 5: Number of experimental conditions that each model fits best, and worst (by item-level correlation).